PHASE III / OBJECTIVES AND OPEN PROBLEMS

AUTONOMOUS FIRE ROBOT

A robot that drives through the house on its own, puts out fires and goes back to its base to refill.

Afonso Teixeira · LETI, ISEP

PHASE I / REMOTE CONTROL
Wi-Fi driving, live video and line following with the ESP32-CAM.

PHASE II / FIREFIGHTER
Flame detection on the Raspberry Pi 5, pan/tilt aiming and water.

PHASE III / AUTONOMOUS
Navigate the house, find and put out fires, refill at the base.

PROJECT DOCUMENTATION / 2026

1. The goal

The robot should move around the house on its own, look for fires and put them out. When its water runs low, it goes back to a water base, docks and refills, and then carries on.

This phase starts from the Phase II robot, which follows a line, stops for a flame, aims and sprays. Its design and results are in the *Engineering Report* (docs/Technical_Report.pdf).

2. The focus of this phase

FOCUS 1

Find and put out fires

- Navigate the house autonomously, without a line on the floor.
- Search for fires along the way.
- Put them out.

FOCUS 2

Refill on its own

- Know when it has to refill.
- Recognise the water base and dock into it.
- The base knows when the robot is ready to receive water.

Everything else in this document supports these two points.

3. Problems to solve

Problems from the previous version, and things that can be better. They are only stated here.

#	PROBLEM	WHAT HAPPENS TODAY
1	**The chassis bends**	Too much weight sits in the middle of the chassis, and it starts to bend. *First option: print it in a stiffer filament, keeping the current layout.*
2	**Two main boards**	The ESP32-CAM is still in the robot. The ESP32 is to be removed, and the Raspberry Pi takes over the parts it controlled that matter for this phase.
3	**The car gets stuck**	The motors get a fixed PWM value and nothing checks that the wheels actually turn. The same 50 % does not do the same with a full and a half-empty battery. *Idea: wheel encoders.*
4	**Too many wires**	The electronics are wired by hand on protoboards. A PCB is wanted. Open question: how wires from parts that are not on the PCB (for example the Raspberry Pi) connect to it. *Idea: header pins on the PCB, like the Pi's GPIO header.*
5	**How the car turns**	The car does not always need to turn on its own axis; it could also turn its wheels. How to build that is still unknown.
6	**Two wheels drag in turns**	To turn left, the right motors drive and the left ones stop, so the stopped wheels drag. All four wheels should work in a turn: one more pin to the L293D drivers.
7	**Water level unknown**	The robot does not know how much water it has. A level sensor in the tank (a water bottle) should tell it when to refill, and a second sensor should tell when the water is about to overflow.
8	**The car shakes**	The car vibrates a lot, and the upper deck that carries the pan/tilt shakes with it.
9	**The camera does not stay steady**	When the car shakes, the camera trembles, which makes flames harder to find. It should keep looking forward, like a tank's gun when the tank climbs an obstacle. *Idea: an IMU.*

Table 3.1 — Problems for Phase III. The solutions are not part of this document.

4. The problems are connected

These problems cannot be solved one at a time in isolation. A change in one area moves the others.

For example, using only the Raspberry Pi means taking out the ESP32. That frees space, so the components may be laid out differently, and a new layout can change:

Remove the ESP32 → **More space** → **New layout** → **Water tank size** **Weight on the structure** **Chassis design** **Steering system** **...**

So the decisions have to be taken together, looking at the robot as one system.

5. What comes next

A second document will follow. It will explain how each problem in Table 3.1 was solved, and how all the areas of the project — mechanics, electronics, software, navigation and the water base — were connected together.

> NOTE This is an experimental candle-scale prototype, not a fire-safety device.